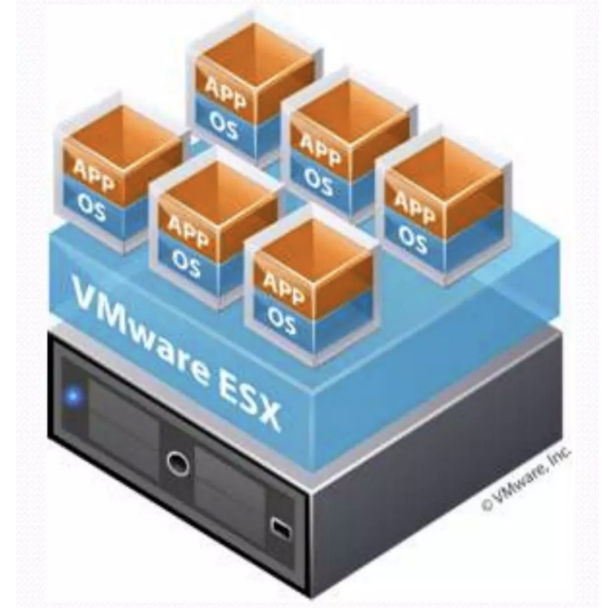


Why Virtualization

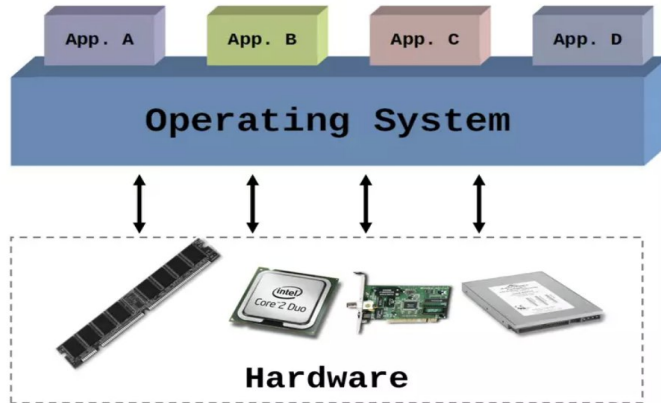
- Multiple Operating systems on a Single System
- Share the underlying Hardware resources
- Server Consolidation
- Simulation of Hardware and Networking Devices
- Application and System mobility
- Virtualization is FUN.... And plenty other reasons

Virtualization

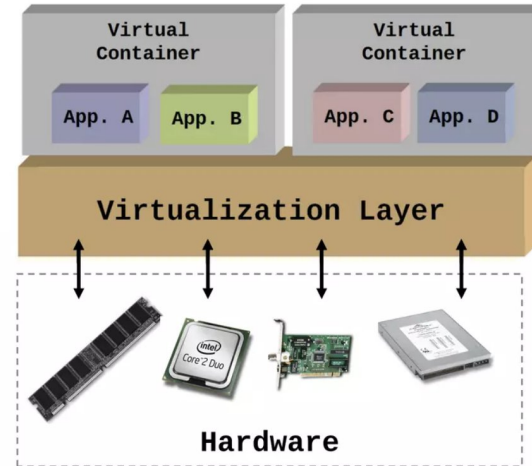
- Virtualization is a Technology that transforms hardware into software
- Virtualization allows to run multiple OS on a Virtual Machine
- Provides a layer of abstraction between computing, storage and networking hardware, and the applications running on it.



Virtualization (Contd.)



'Non virtualized' system A single OS controls all hardware platform resources



Virtualized system. It makes it possible to run multiple Virtual Containers on a single physical platform

About Virtual Machines

A virtual machine (VM) is a software representation of a physical computer and its components

Virtual Machine

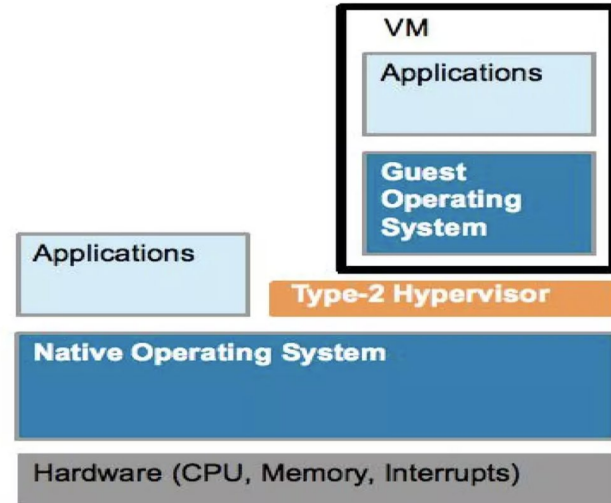
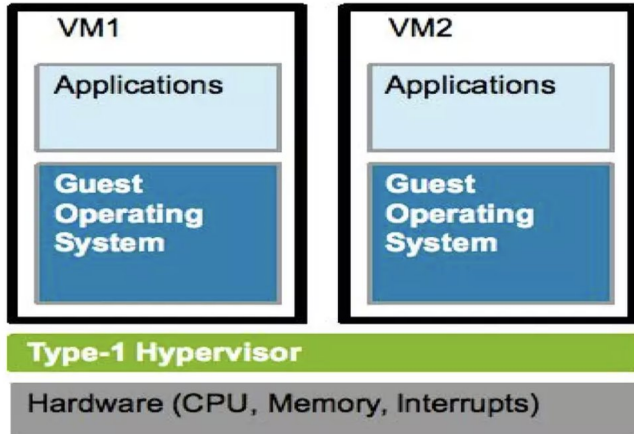


The virtual machine includes the following components:

- Guest OS
- Vmware tools (In case of ESXi hypervisor)
- Virtual resources, such as:
 - CPU and memory
 - Network adapters
 - Disks and controllers
 - GPUs

Note: VMware Tools is a bundle of drivers. Other Hypervisors also have such kind of drivers using which the guest OS can interact efficiently with the Virtual hardware

What is a Hypervisor?

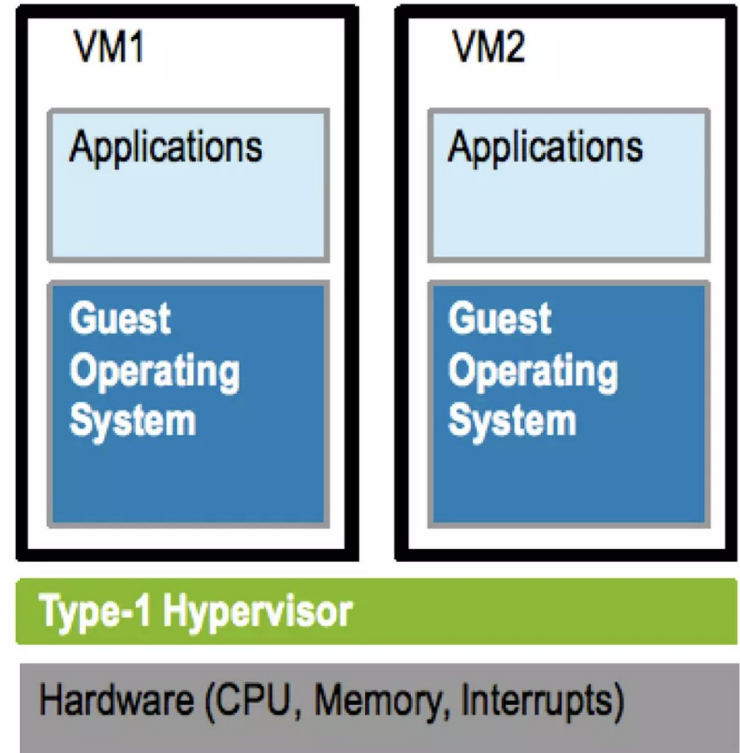


1. A hypervisor is a piece of computer software, firmware or hardware that creates and runs virtual machines. (i.e virtual machine manager or virtual machine monitor).
2. **Host machine:** The computer that runs and has one or more virtual machines.
3. Each virtual machine is called a **guest machine**

Classification of Hypervisor

Type 1 Hypervisor

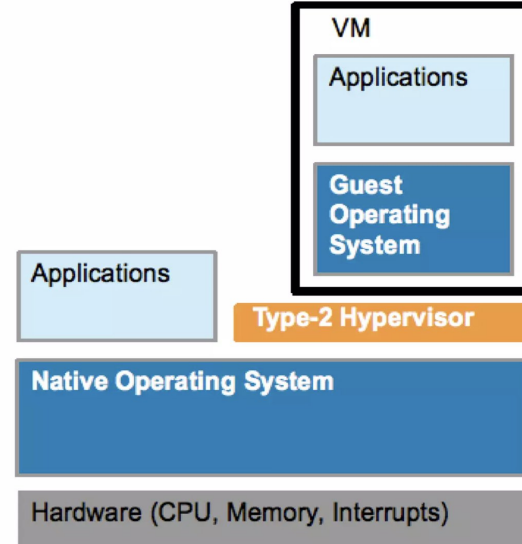
- Runs directly on the host hardware to manage guest operating systems
- Does not require any base server operating system
- Direct access to hardware resources
- Better performance, scalability, and stability
- However, hardware support is limited



Classification of Hypervisors

Type 2 Hypervisor

- Hosted on the main operation system
- Basically a software installed on an OS
- Hypervisor asks OS to make hardware calls
- Better compatibility with hardware
- Increased overhead affects performance



Virtualization Techniques

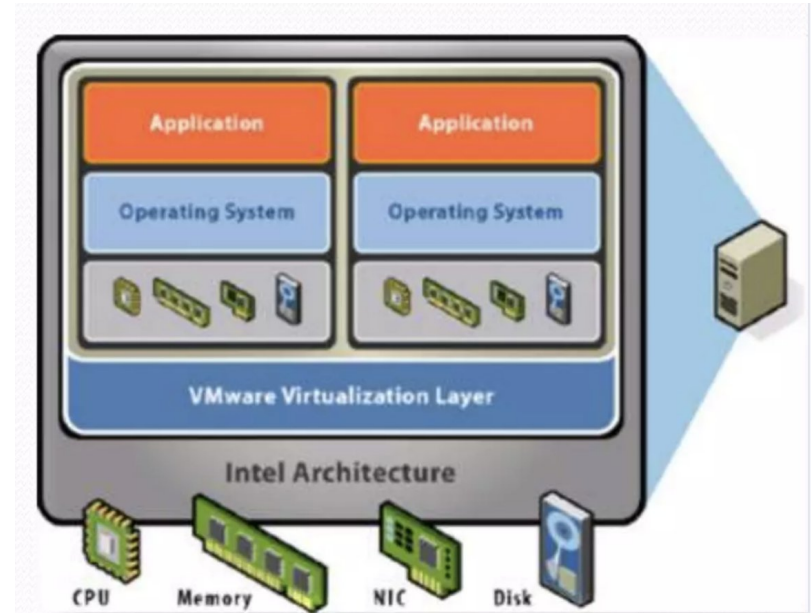
There are multiple approaches to running the guest operating system. These include:

- Full Virtualization
- Paravirtualization
 - Also Known as OS assisted virtualization
- Hardware-assisted virtualization
 - Also known as accelerated virtualization, hardware virtual machine

Virtualization Techniques

Full Virtualization

Full virtualization uses a special kind of software called a hypervisor. The hypervisor interacts directly with the physical server's CPU and disk space. The servers as a platform for the virtual server's operation systems



Virtualization Techniques

Paravirtualization

The para-virtualization approach is a little different than the full virtualization technique where the guest servers in a para virtualization systems are aware of each other

Virtualization Techniques

Hardware-assisted virtualization

- Enables efficient full virtualization using help from hardware capabilities, primarily from the host processors
- Unmodified guest OS; no API calls made
- Hypervisor traps sensitive calls

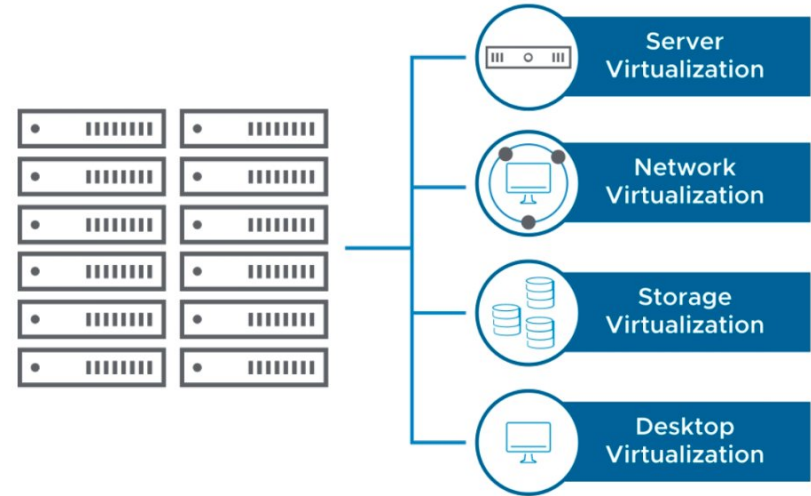
Types of Virtualization

Server virtualization allows companies to create multiple operating systems that run on a single physical server, called VMs. Each VM has access to the underlying server's computing resources.

Network virtualization is the complete reproduction of a physical network in software. Applications run on the virtual network exactly as if on a physical network.

Storage virtualization is the process of creating a software-based representation of network storage devices into what appears to be a single unit.

By deploying desktops as a managed service, you can respond more quickly to changing needs and opportunities



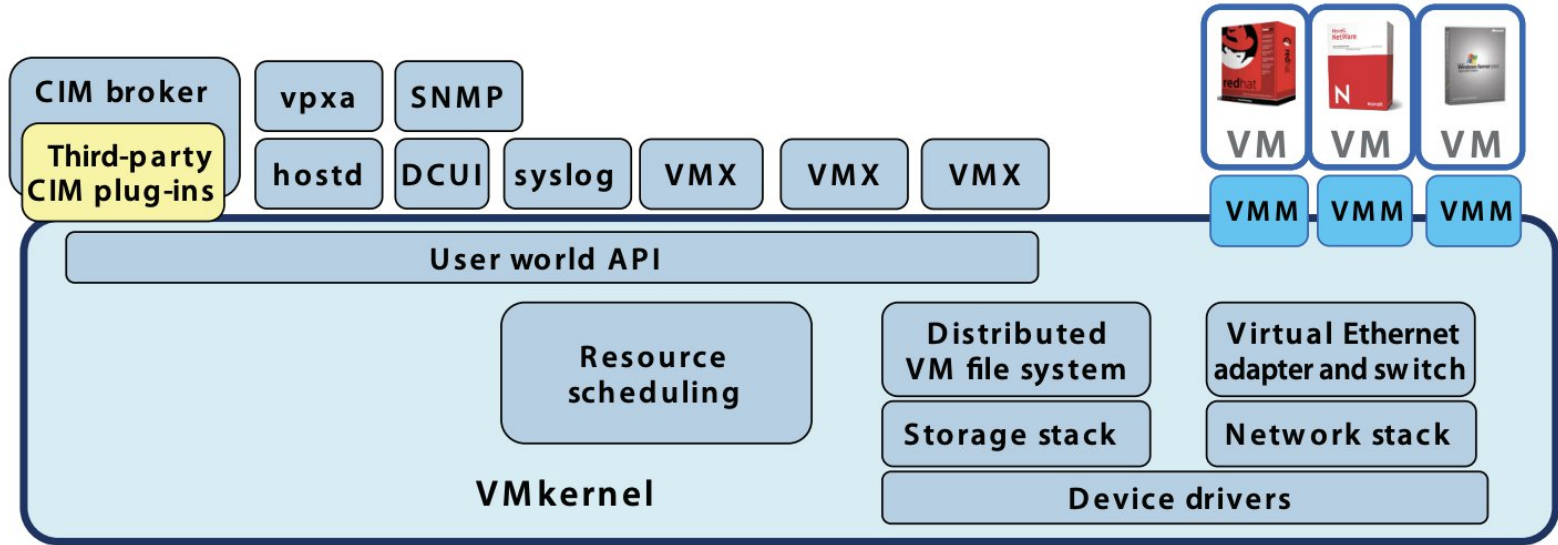
ESXi architecture

Vmware ESXi is the next-generation hypervisor providing a new foundation for virtual infrastructure.

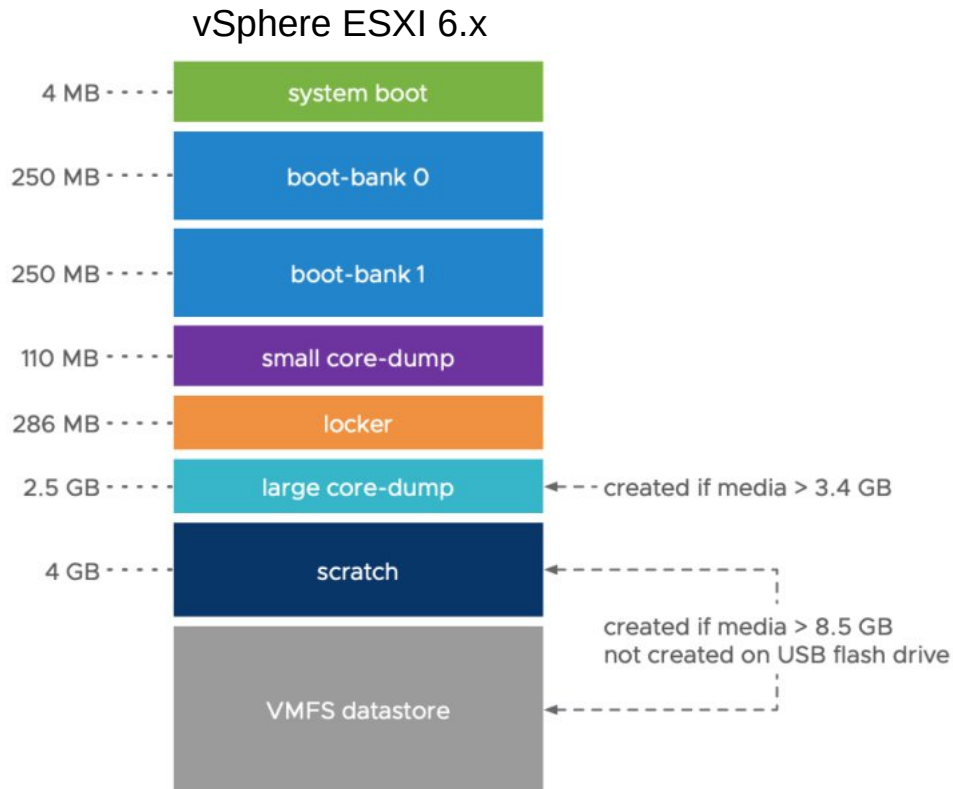
The VMware ESXi architecture comprises

- The underlying operating system, called VMkernel, and processes that run on top of it. VMkernel provides means for running all processes on the system, including management applications and agents as well as virtual machines. It has control of all hardware devices on the server, and manages resources for the applications.
- DCUI – Interface accessible through the console of the server
- VMM – Process that provides the execution for a virtual machine. Each VM has its own VMM and vmx process
- CIM (Common Information Model) – Interface that enables hardware-level management from the remote application via a set of standard APIs

Architecture Diagram



ESXi partition Layout

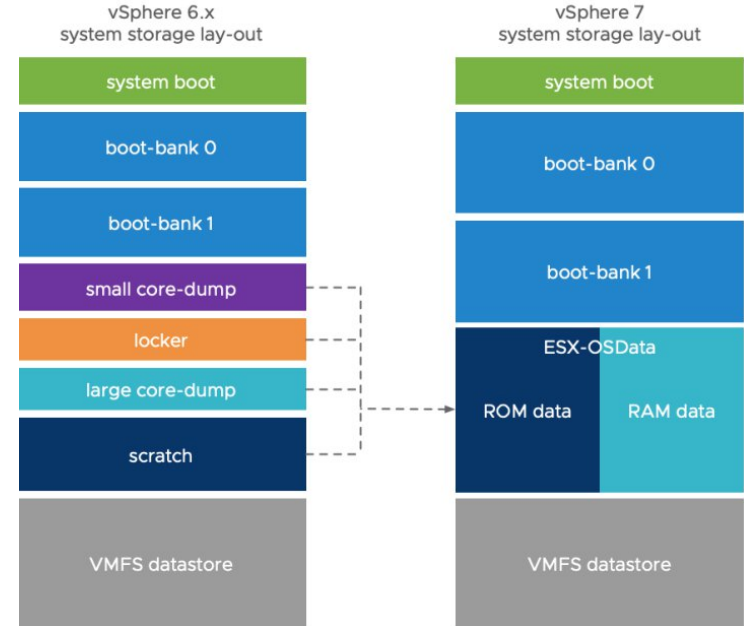


Difference in partition layout

ESXi 7 system storage lay-out consists of 4 partitions

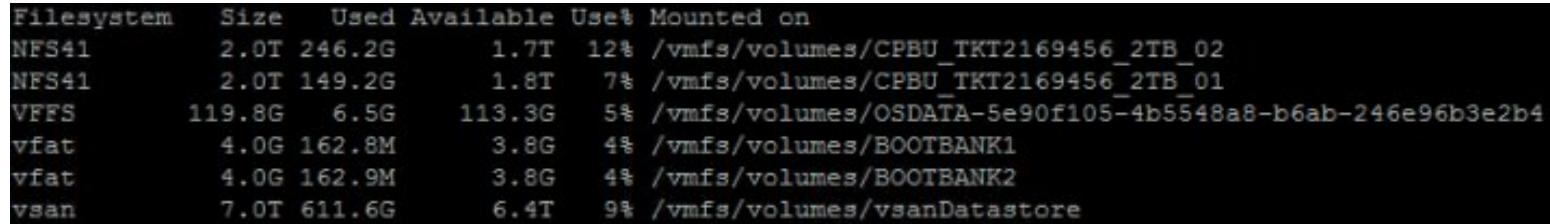
- System boot
 - Stores boot loader and EFI modules.
 - Type: FAT16
- Boot-banks
 - System space to store ESXi boot modules
 - Type: FAT16
- ESX-OSData
 - Acts as the unified location to store extra (nonboot) modules, system configuration and satet and system virtual machines
 - Type: VMFS-L

The OSData partition is divided into two high-level categories of data called ROM-data and RAM-data.



Review the System Storage Lay-out

You can use CLI of an ESXi host which looks like below image to

A screenshot of an ESXi host's command-line interface showing the output of a storage-related command. The output is a table with columns: Filesystem, Size, Used, Available, Use%, and Mounted on. The rows represent different storage volumes: two NFS41 volumes, a VFFS volume (OSDATA), two vfat volumes (BOOTBANK1 and BOOTBANK2), and a vsan volume (vsanDatastore).

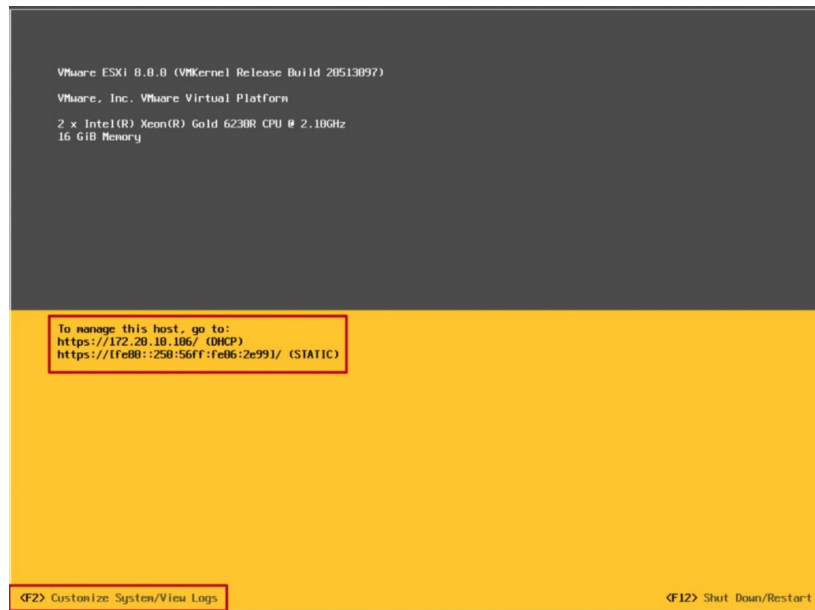
Filesystem	Size	Used	Available	Use%	Mounted on
NFS41	2.0T	246.2G	1.7T	12%	/vmfs/volumes/CPBU_TKT2169456_2TB_02
NFS41	2.0T	149.2G	1.8T	7%	/vmfs/volumes/CPBU_TKT2169456_2TB_01
VFFS	119.8G	6.5G	113.3G	5%	/vmfs/volumes/OSDATA-5e90f105-4b5548a8-b6ab-246e96b3e2b4
vfat	4.0G	162.8M	3.8G	4%	/vmfs/volumes/BOOTBANK1
vfat	4.0G	162.9M	3.8G	4%	/vmfs/volumes/BOOTBANK2
vsan	7.0T	611.6G	6.4T	9%	/vmfs/volumes/vsanDatastore

You might notice the OSDATA partition being formatted as the Virtual Flash File System (VFFS). When the OSDATA partition is placed on a SDD or NVMe device, VMFS-L is labeled as VFSS

Configuring an ESXi Host

You can use the ESXi host's DCUI to configure certain settings, such as the host's network settings. The DCUI is a text-based user interface with keyboard-only interaction.

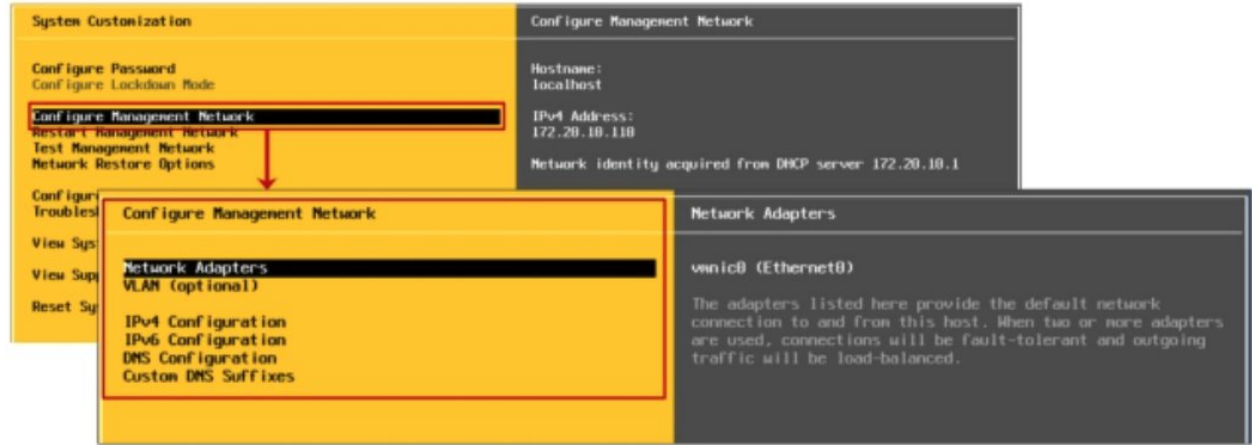
The DCUI is a low-level configuration and management interface, accessible through the console of the server, that is used primarily for initial basic configuration. You press F2 to start customizing system settings.



Contd...

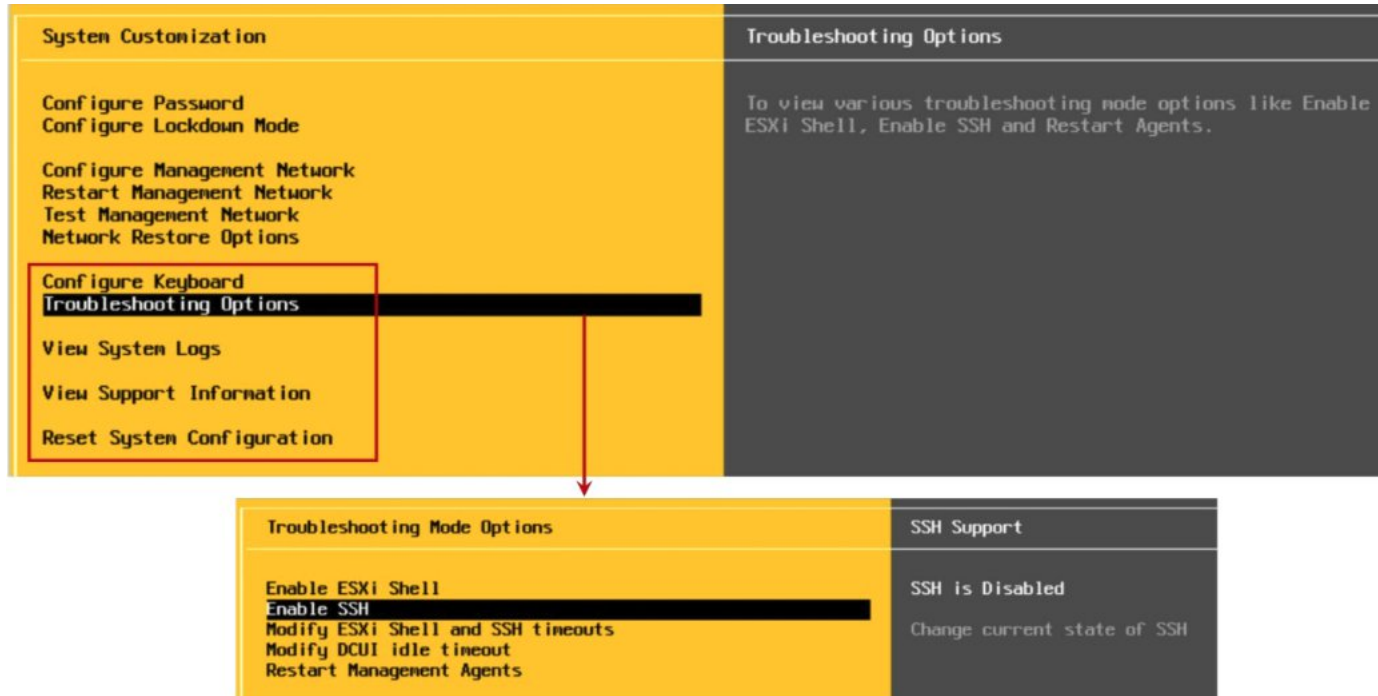
Use the DCUI to configure management network settings:

- Network adapter selection
- VLAN ID
- IPv4 and IPv6 configuration
- Host name
- DNS servers and suffixes



Contd...

DCUI can be used to configure other settings like keyboard layout, activate troubleshooting services, view support information, and view system logs.



NUTANIX

THANK YOU

